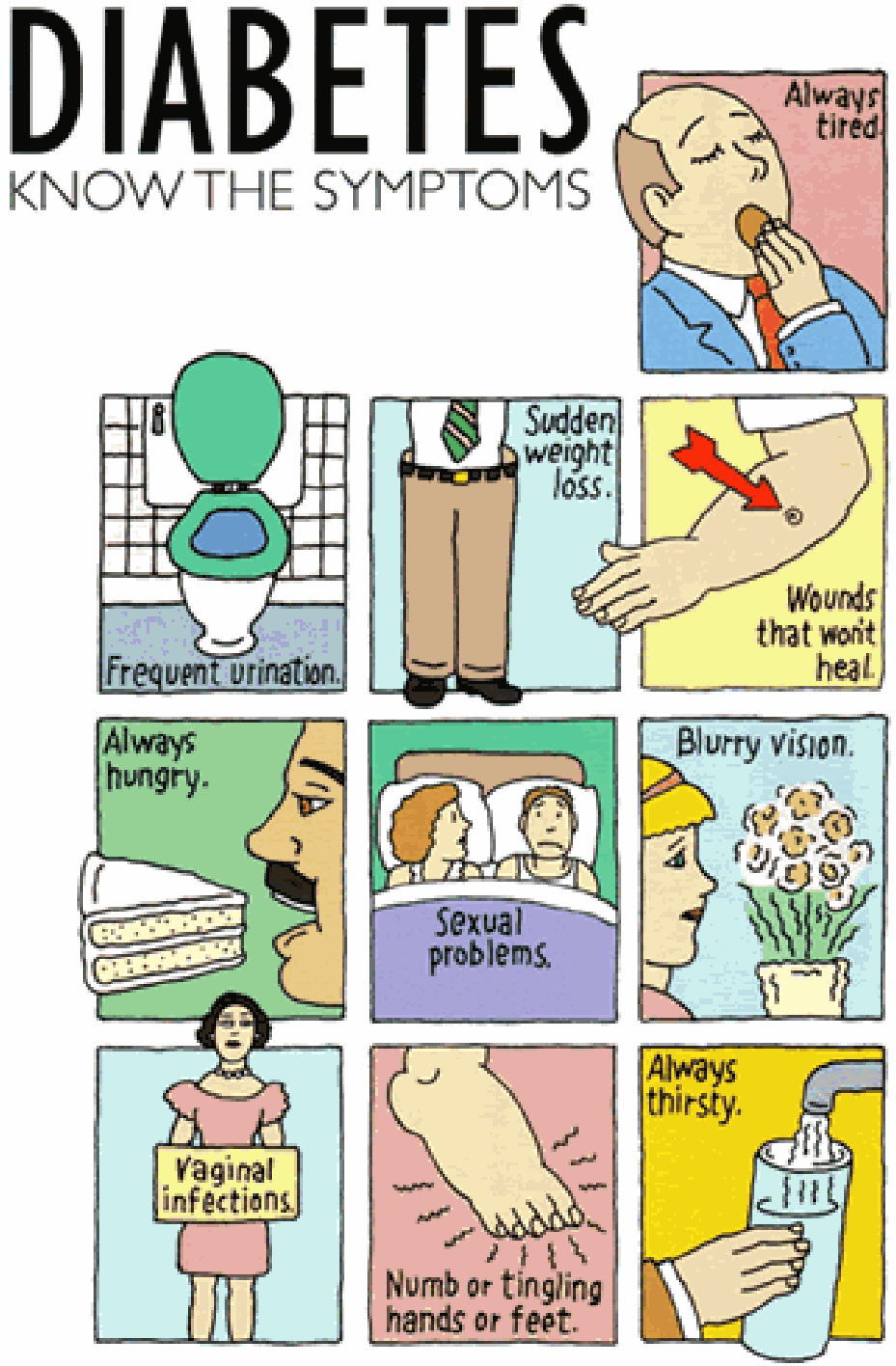



| Diabetes mellitus | Diabetes insipidus |
|---|---|
| It results from hyposecretion of insulin. | It results from hyposecretion of ADH. |
| Excretion of urine with sugar. | Excretion of large amounts of dilute urine. |
| Excessive eating. | Dehydration. |

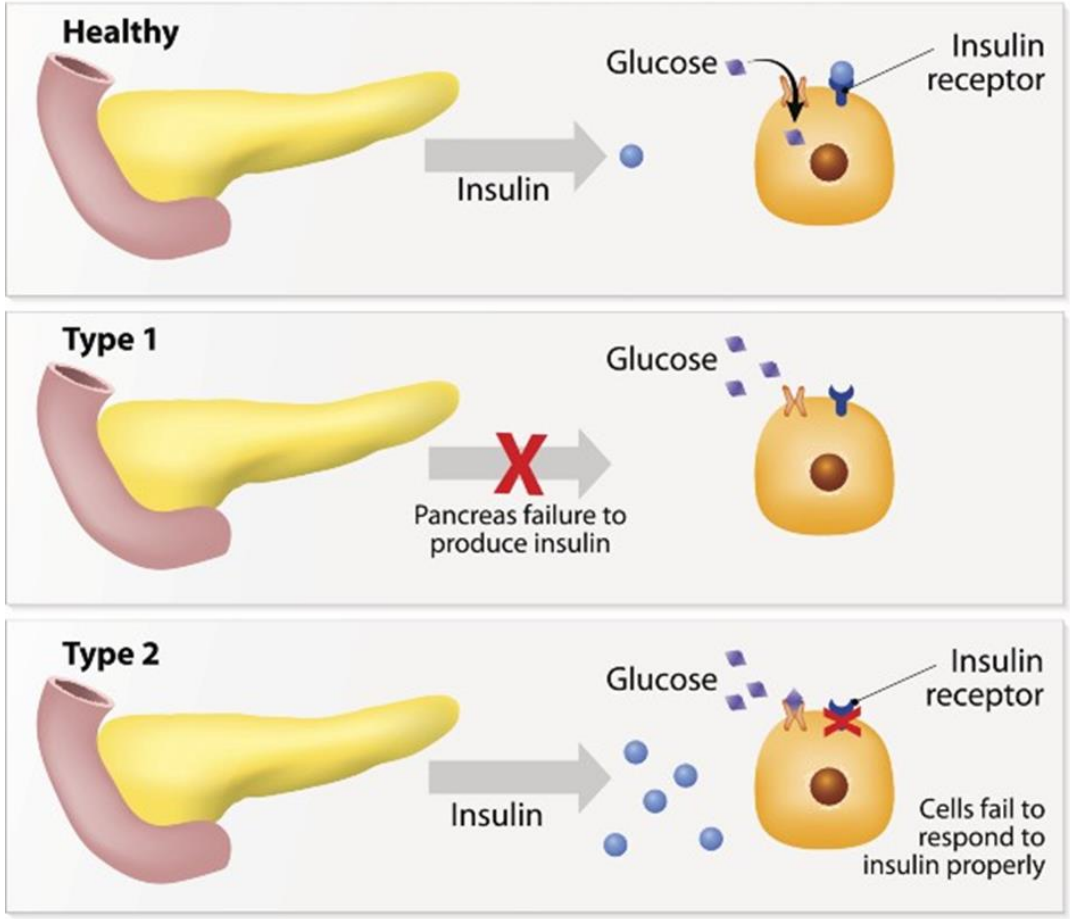
# DIFFERENCE BETWEEN DIABETES MELLITUS AND DIABETES INSIPIDUS

Diabetes insipidus should not be confused with diabetes mellitus (DM), which results from insulin deficiency or resistance leading to high blood glucose, also called blood sugar.

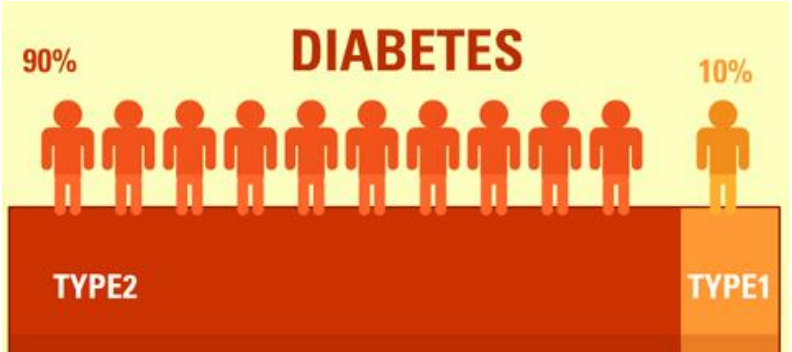
Diabetes insipidus and diabetes mellitus are unrelated, although they can have similar signs and symptoms, like excessive thirst and excessive urination.

Diabetes mellitus is far more common than diabetes insipidus and receives more news coverage. Diabetes mellitus has two main forms, type 1 diabetes and type 2 diabetes. Diabetes insipidus is a different form of illness altogether.

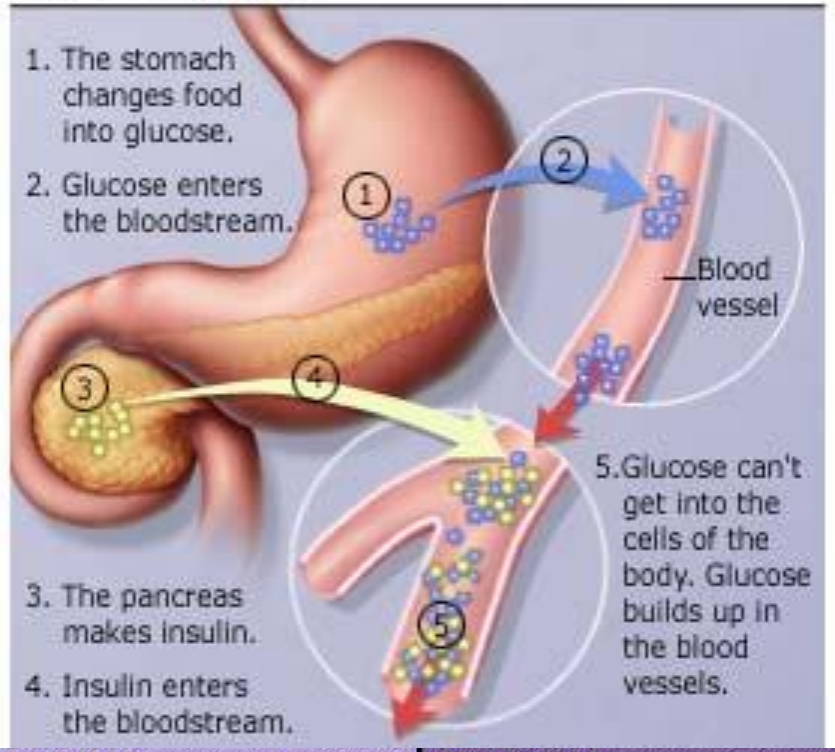
# DIABETES MELLITUS

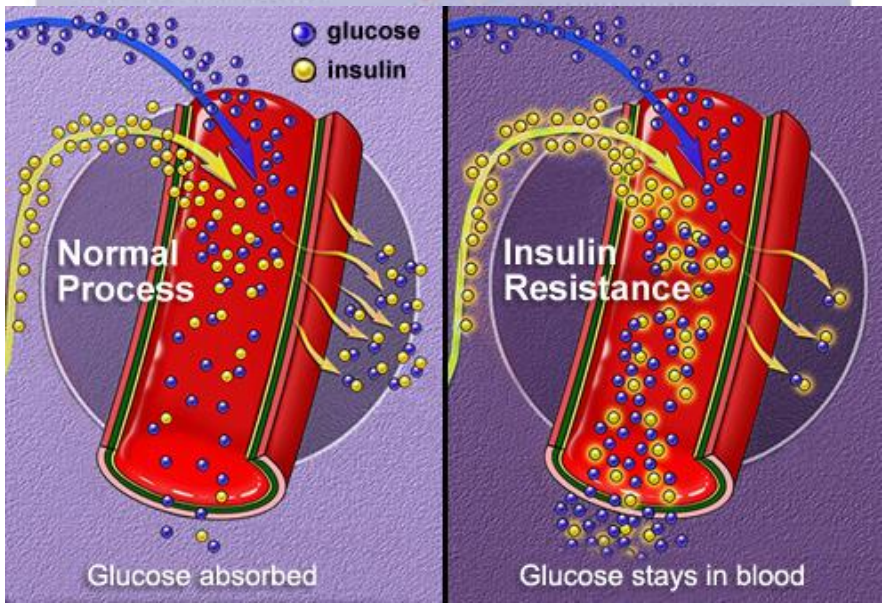



| Type 1 | Type 2 |
|---|---|
| Usually young | Usually older |
| Short history- acute onset | Insidious onset |
| Not overweight | Often overweight |
| Insulin deficiency | Insulin resistance |
| Rare | Common |
| Requires insulin from diagnosis | Diet and lifestyle change can reverse it<br>Then add oral medications<br>May require insulin |
| Often random | Strong family history |

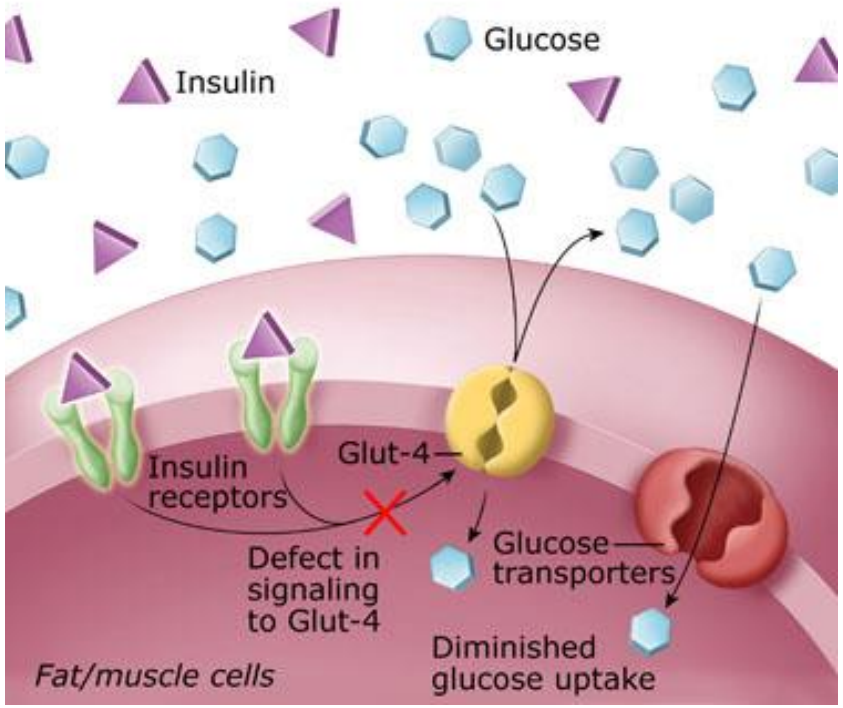


## Type 2 Diabetes

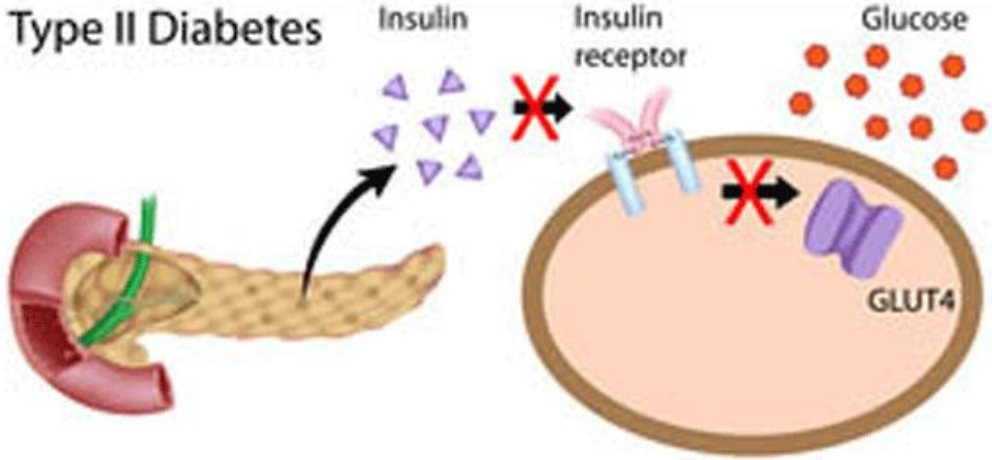


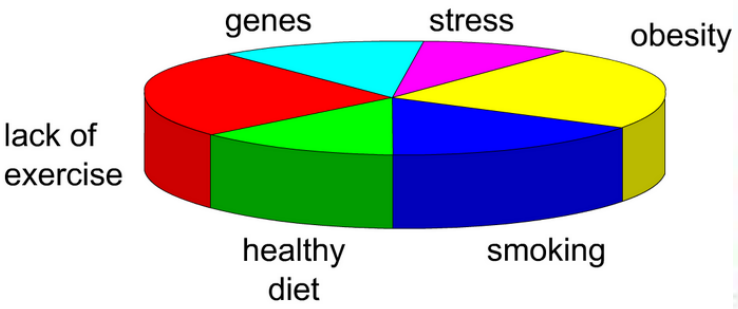




## Type 2 Diabetes: Insulin Resistance

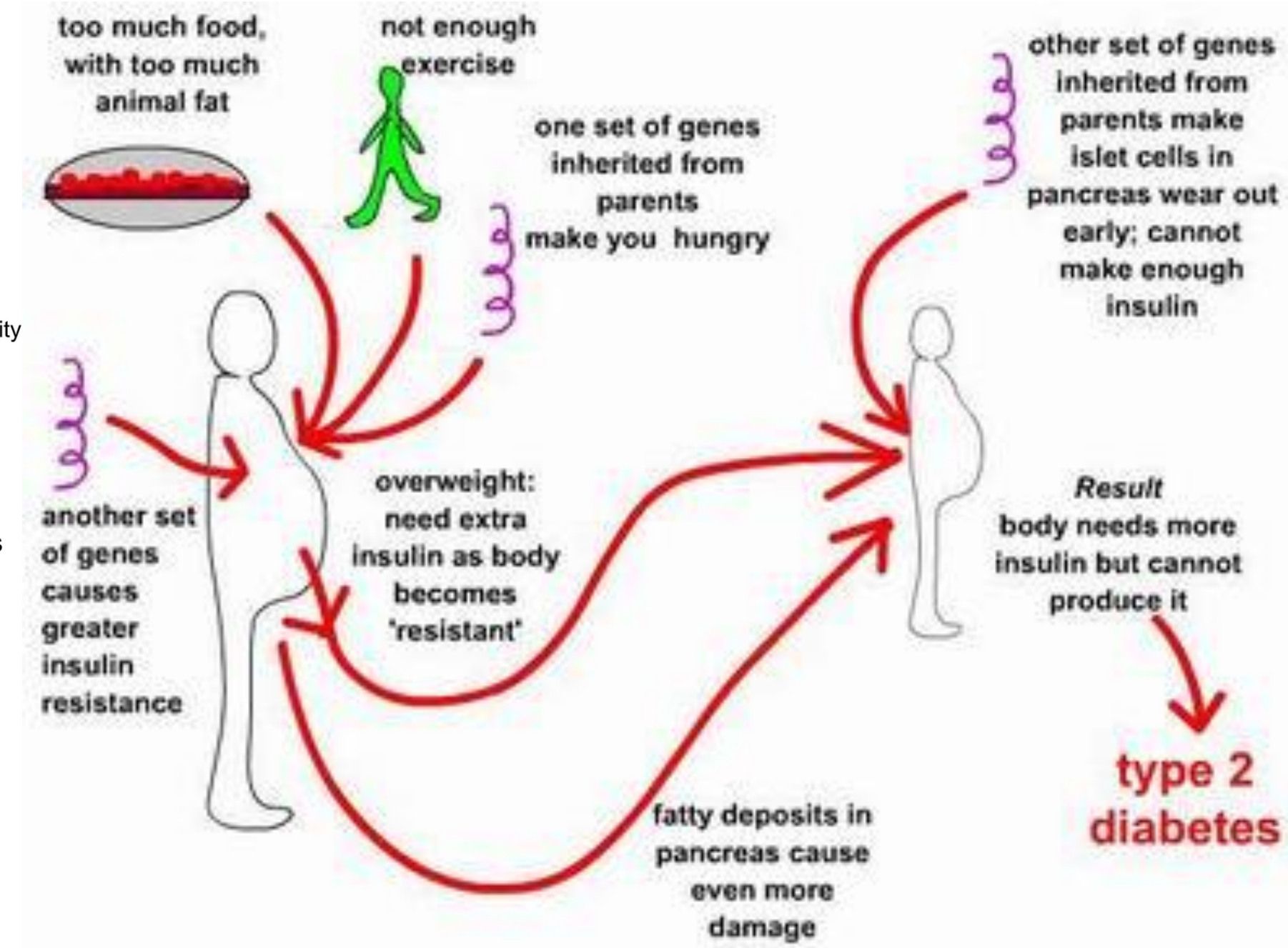



## Pancreas and Type II Diabetes






factors contributing to type 2 diabetes

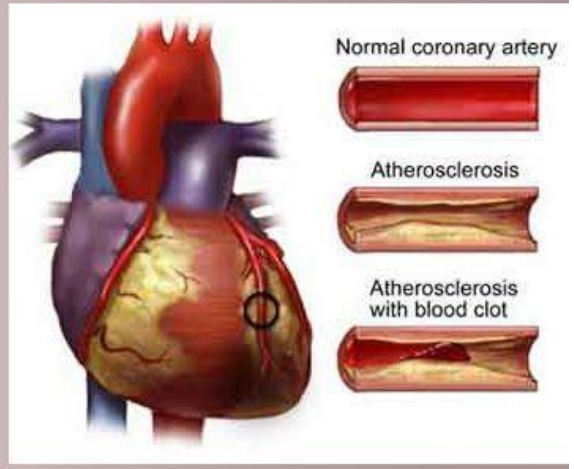


# Types Of Cardiovascular Disease

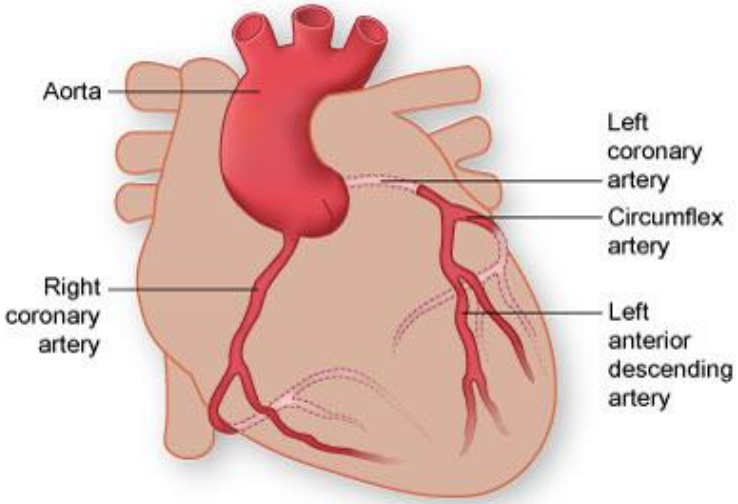
- Atherosclerosis ----(Plaque )
- Coronary heart disease (CHD) ---Myocardial infarction (MI) or heart attack
- Chest pain (angina pectoris)
  - Ischemia (an inadequate blood and oxygen supply to the heart muscles.)
- Irregular heartbeat (arrhythmia)
  - » Tachycardia = more than 100 beats/min
  - » Bradycardia = less than 60 beats/min
- Congestive heart failure (CHF)
  - heart muscle is unable to keep blood circulating normally
- Congenital heart disease (problem in the structure of the heart that is present at birth)
- Stroke--- Myocardial infarction (MI)
- **Embolism**: blockage of blood vessels

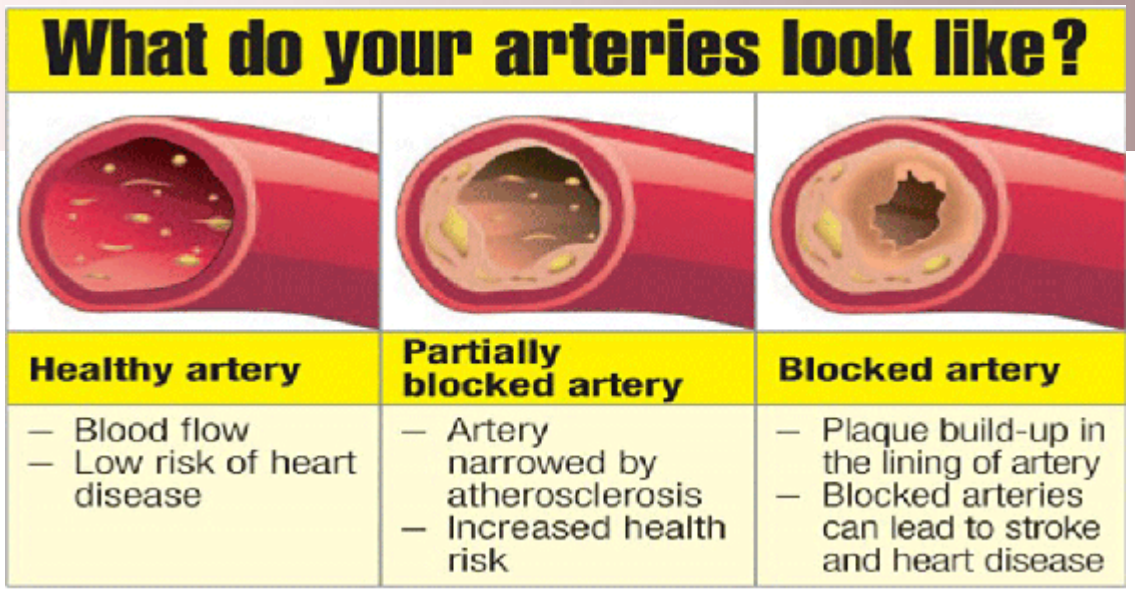
# Coronary Heart Disease

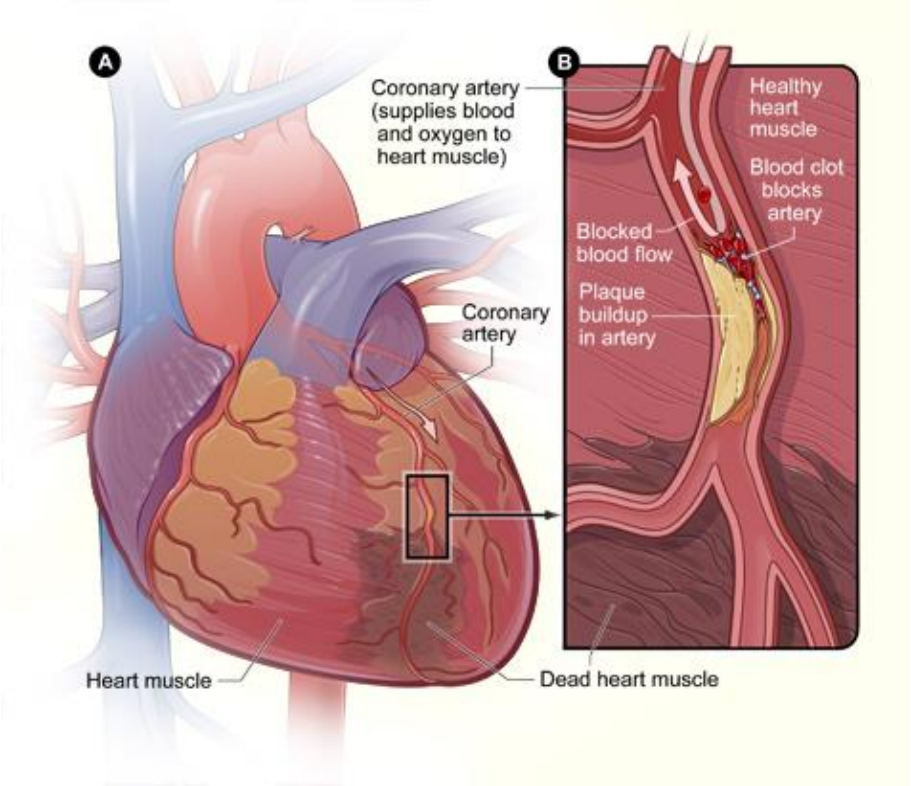
- Slow build up of plaque (lipids, cholesterol) = **ATHEROSCLEROSIS**
- Arteries become harder, less flexible
- Less space for blood
- Coronary arteries supply $O_2$ to heart cells
- Thrombosis = clot = heart attack = heart cells die




- Factors affecting <u>coronary heart disease</u> (CHD):
  Age, race, heredity, gender, cholesterol levels, blood pressure, obesity, diabetes, smoking, sedentary lifestyle, stress

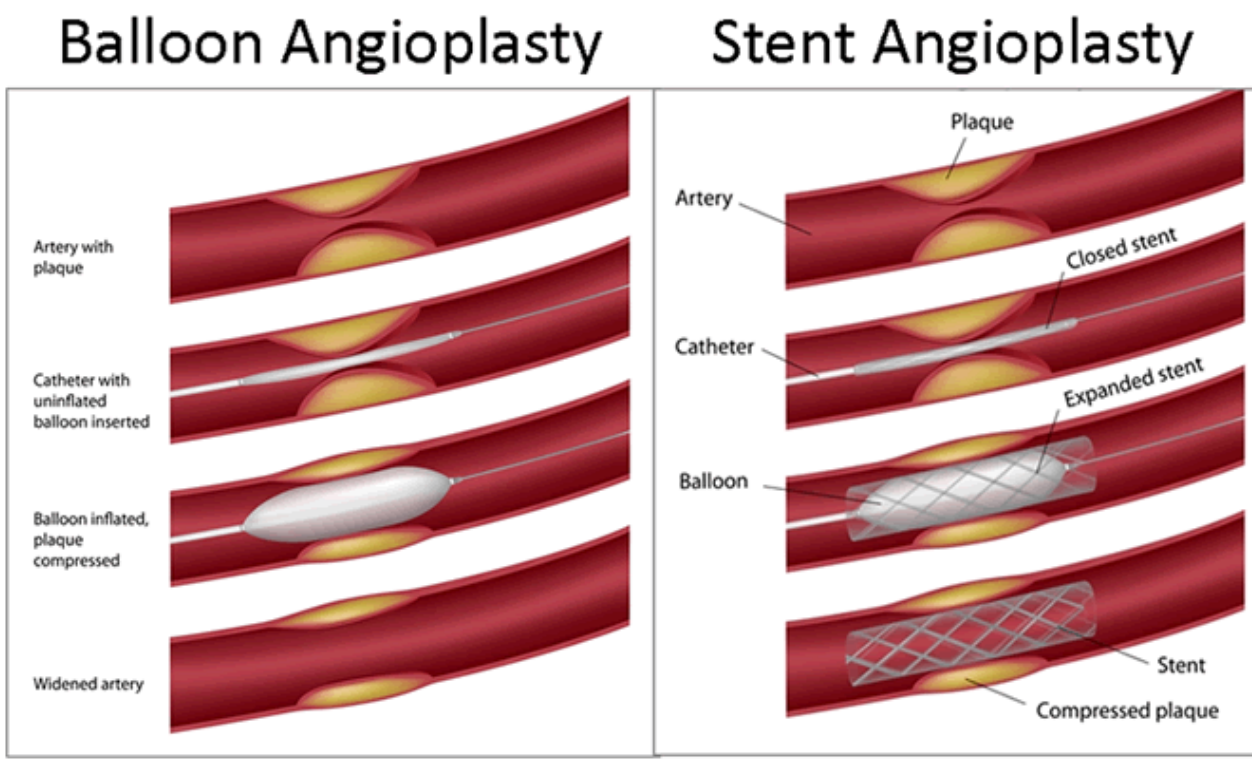


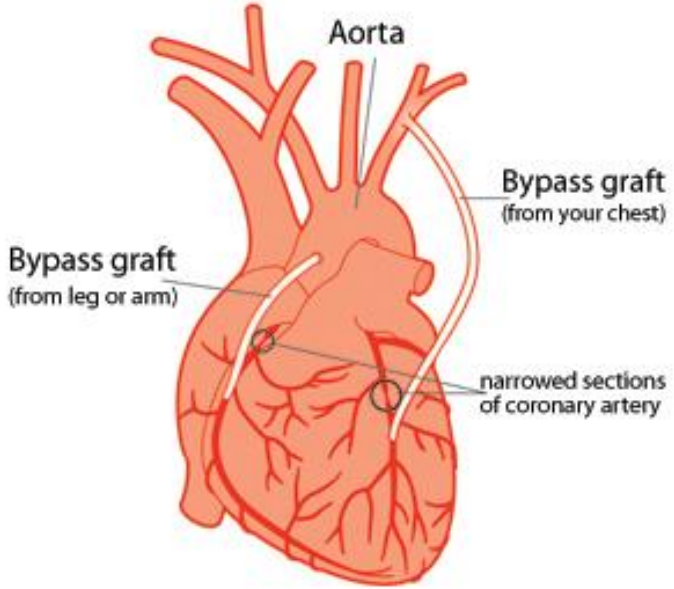



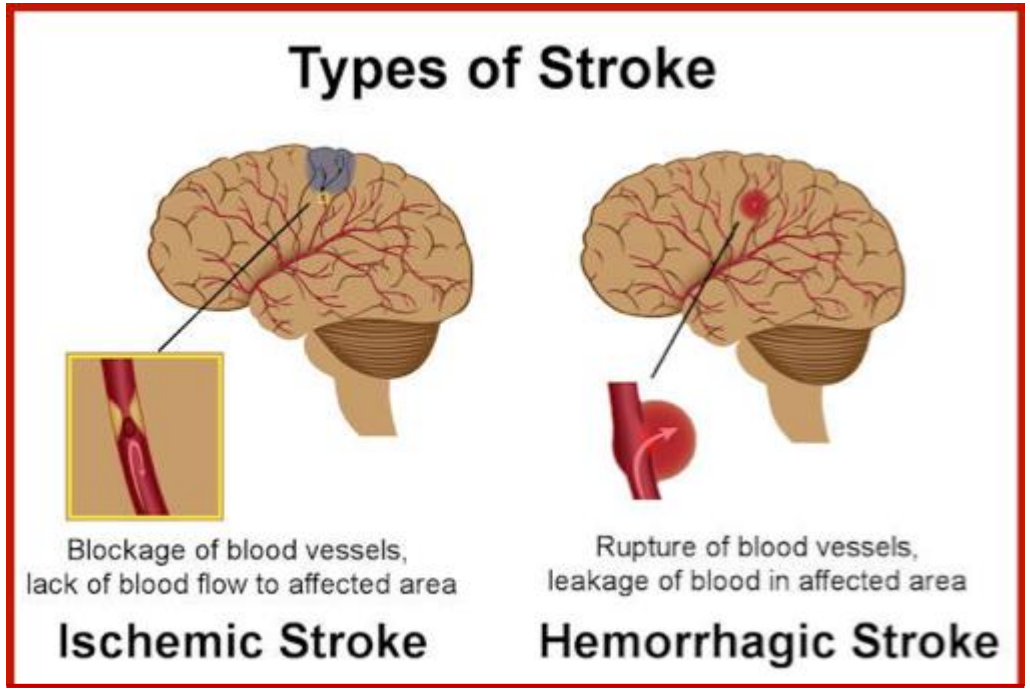
# TREATMENT

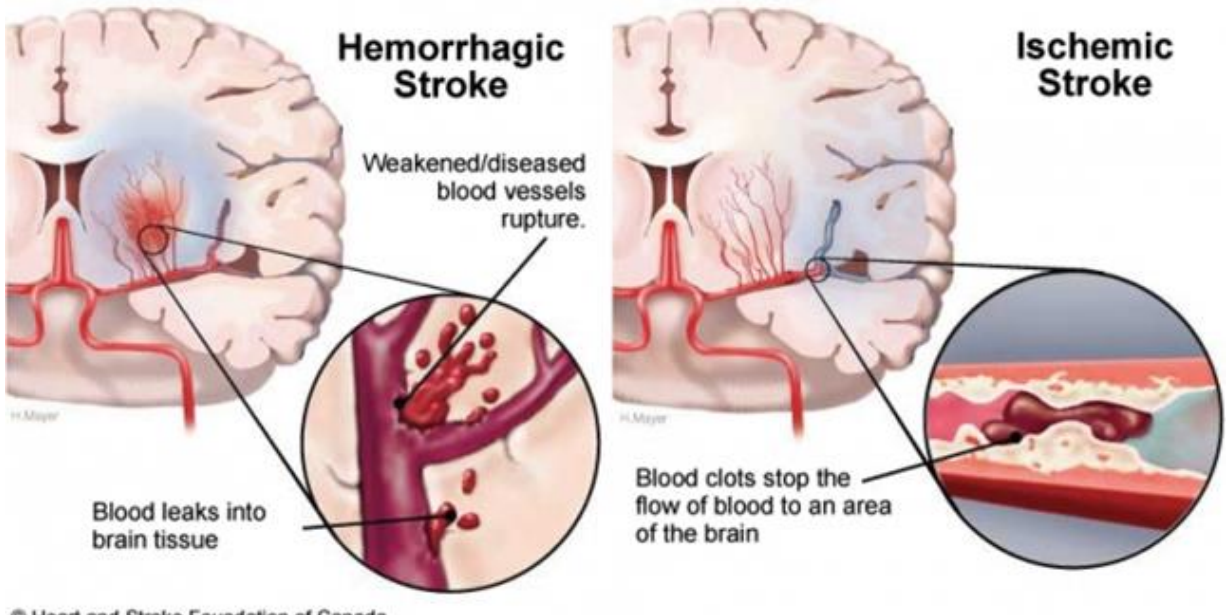
- Treatment of cardiovascular disease depends on the specific form of the disease in each patient, but effective treatment always includes preventive lifestyle changes discussed above.
- Medications, such as blood pressure reducing medications, aspirin and the statin cholesterol-lowering drugs may be helpful.
- In some circumstances, surgery or angioplasty may be warranted to reopen, repair, or replace damaged blood vessels.




Coronary Bypass Surgery



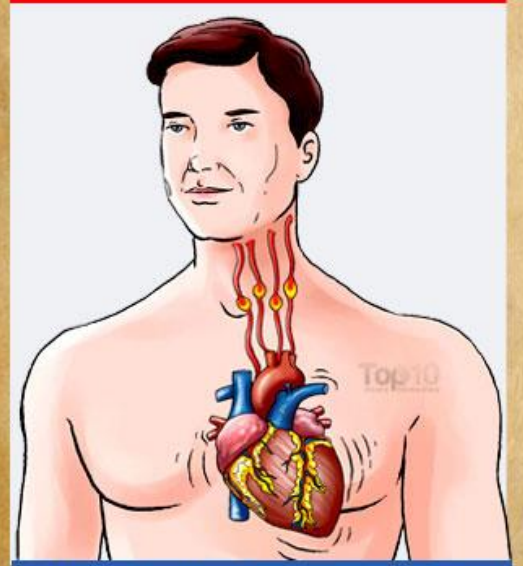




# What is Stroke?

- A **stroke** happens when a blood vessel that carries oxygen to the brain gets blocked or bursts. When that happens, blood can't deliver oxygen to part of the brain and that part of the brain starts to die.
- Without the right medical attention, a stroke can cause major damage to the brain. People who have a stroke may have problems speaking, seeing or moving normally for a short time or for the rest of their lives. The person can even die from a stroke.
- Older adults are more likely to have a stroke than younger people, but people can have a stroke at any age. Other factors that increase risk of stroke include high blood pressure, cigarette smoking, diabetes, other types of cardiovascular disease, and high blood cholesterol.

# YOU MUST KNOW THE DIFFERENCES BETWEEN

## HEART ATTACK

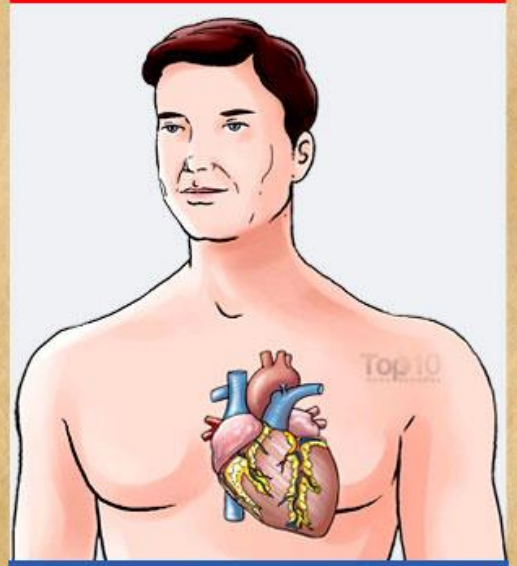


A **heart attack** is a circulation disorder.

SYMPTOMS

- CHEST PAIN (ANGINA)
- BODY ACHES
- SHORTNESS OF BREATH
- COLD SWEATS

## CARDIAC ARREST

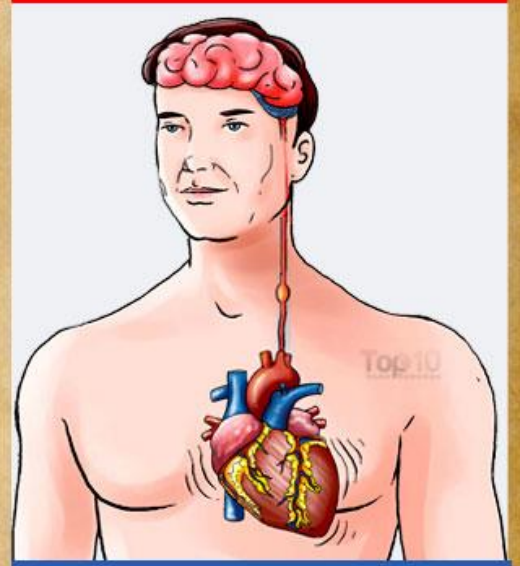


A **cardiac arrest** is an "electrical" disorder.

SYMPTOMS

- LOSS OF CONSCIOUSNESS
- BLACKOUT
- CHEST PAIN
- EXTREME PALPITATION

## STROKE



A **stroke** is a brain disorder.

SYMPTOMS

- MENTAL CONFUSION
- DISRUPTED SPEECH
- INABILITY TO WALK
- BLURRED VISION

**Heart Attack** occurs when blood supply to the heart is **severely reduced or cut off**. Lack of oxygen can **damage the heart** and it will begin to die.

**Cardiac Arrest** occurs when the heart suddenly **stops beating**, **preventing blood and oxygen** from getting to **vital organs**.

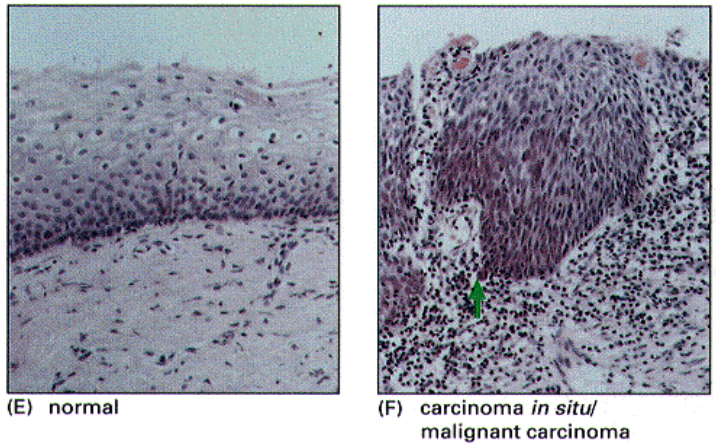
**Stroke** happens when the **blood flow to the brain is interrupted** or severely limtied cutting off oxygen and nutrients from brain tissue. **Brain cells start dying minutes after a stroke.**

Remember, you can stop heart disease before it starts. Start small and keep it simple. Make one change today and then you're ready to make another!

**Life's Simple 7**

1. Avoid smoking and using tobacco products
2. Be physically active every day
3. Eat a heart-healthy diet
4. Keep a healthy weight
5. Keep your blood pressure healthy
6. Keep your total cholesterol healthy
7. Keep your blood sugar healthy

# What is cancer?




From The Art of MBoC³ © 1995 Garland Publishing, Inc.

- Caner is defined as the continuous uncontrolled growth of cells.
- A tumor is a any abnormal proliferation of cells.
- **Benign tumors** stays confined to its original location
- **Malignant tumors** are capable of invading surrounding tissue or invading the entire body
- Tumors can arise from any cell type in the body

Abnormal cell growth (**neoplasia**)

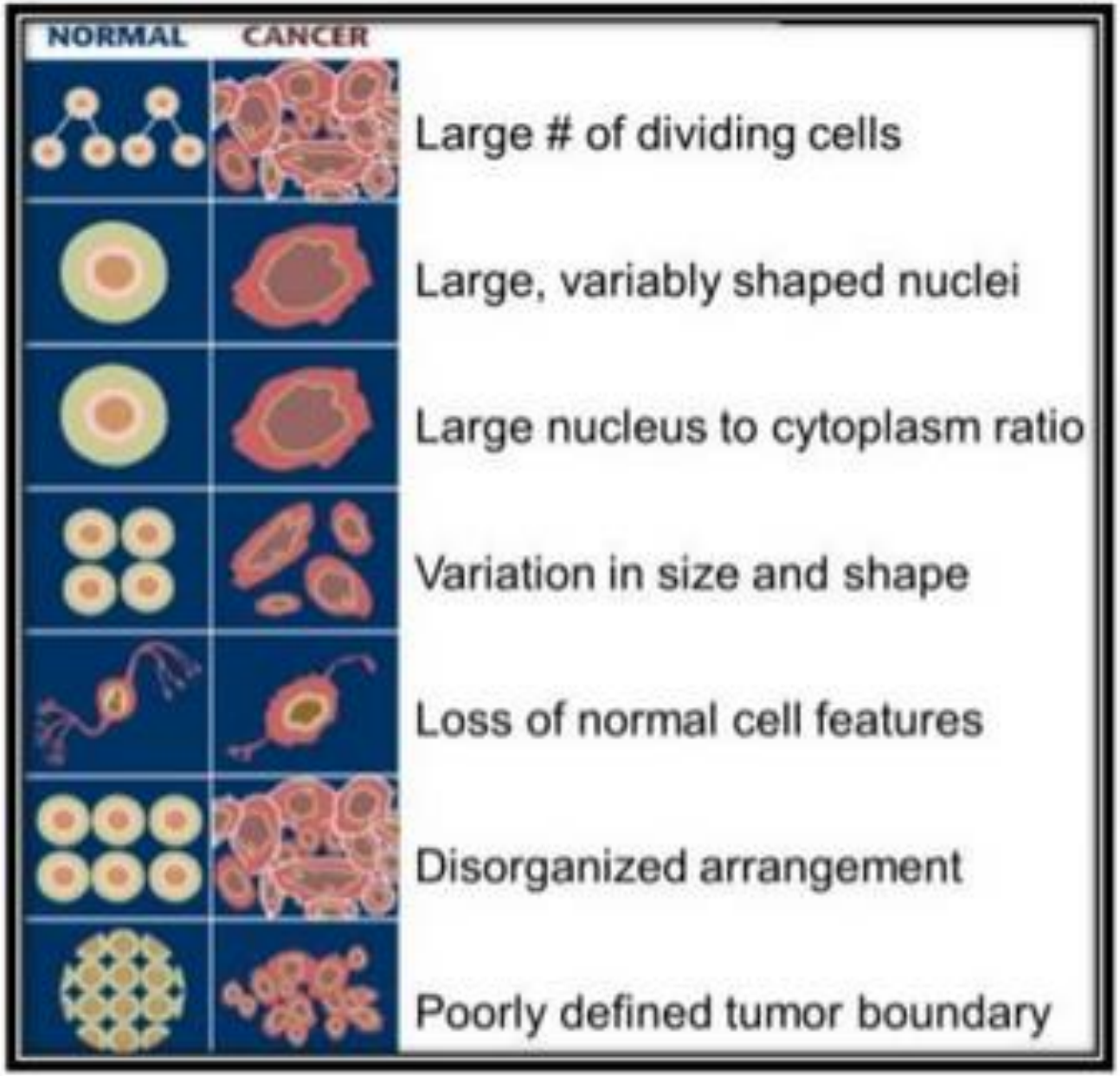
**Benign:** slow growth, non-invasive, no metastasis

**Malignant:** rapid growth, **invasive**, potential for metastasis

## Is cancer a heritable disease?

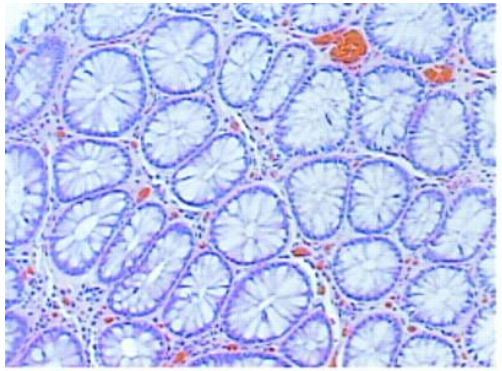
There are heritable cancer syndromes. The majority of cancers, however, are not familial. Cancer is a genetic disease, but the majority of mutations that lead to cancer are **somatic**

# Normal Cells vs. Cancer Cells

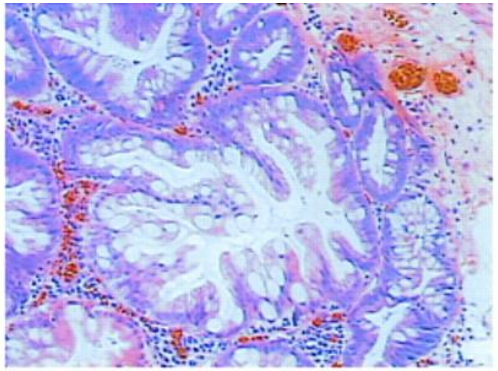


# The Vocabulary

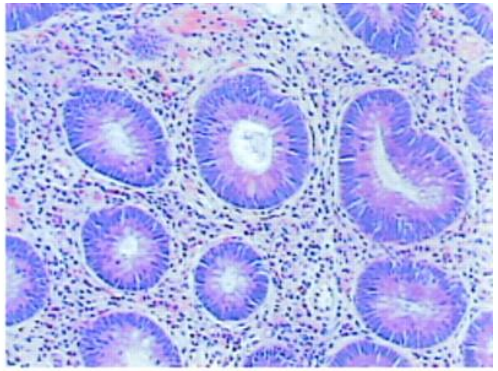
- **Hyperplasia** – increased number of cells
- **Hypertrophy** – increased size of cells
- **Dysplasia** – disorderly proliferation
- **Neoplasia** – abnormal new growth
- **Anaplasia** – lack of differentiation
- **Tumor** – originally meant any swelling, but now equated with neoplasia
- **Metastasis** –growth at a distant site



Normal



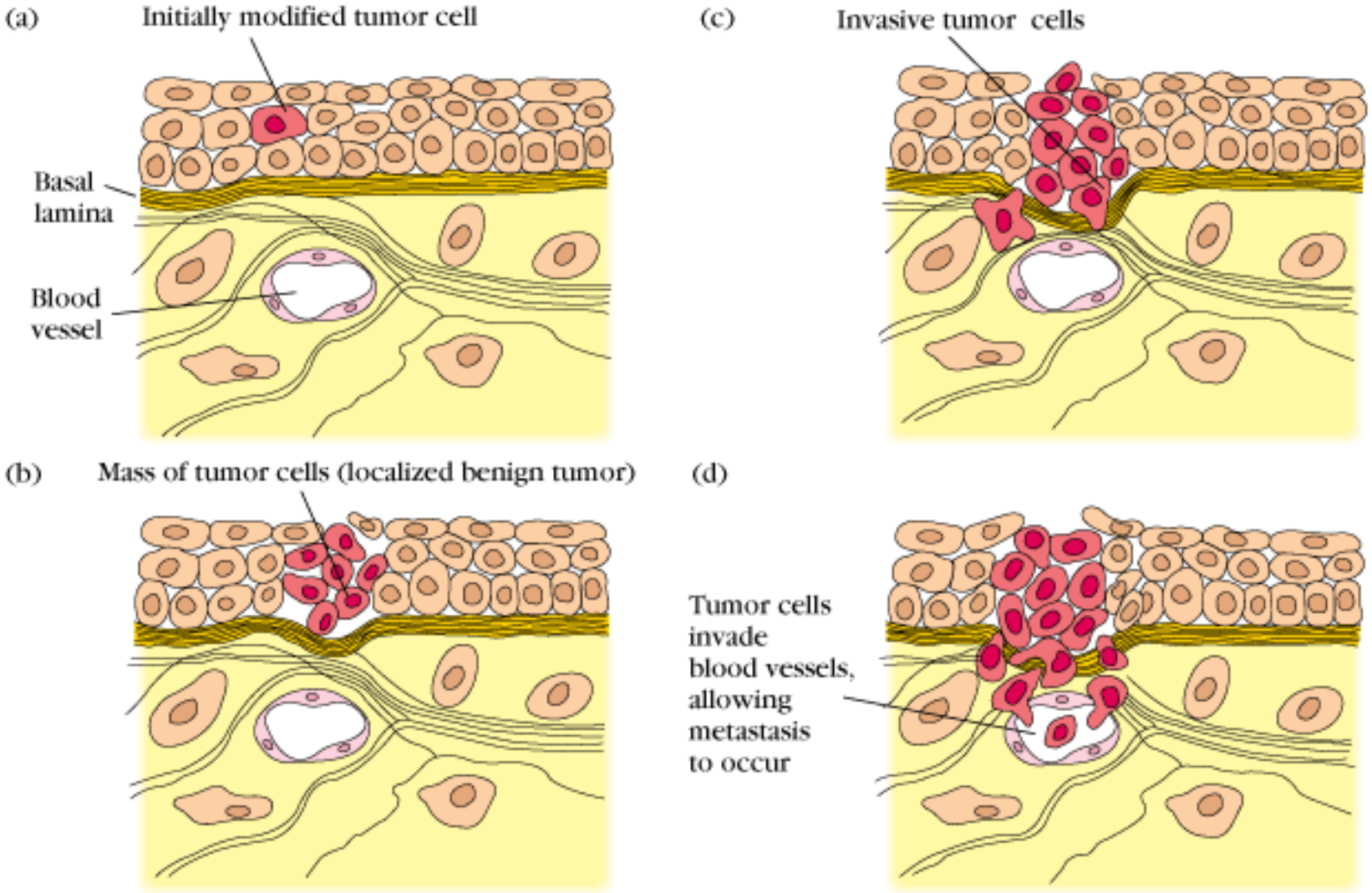
Hyperplastic



Dysplastic

# Most malignant cells become metastatic

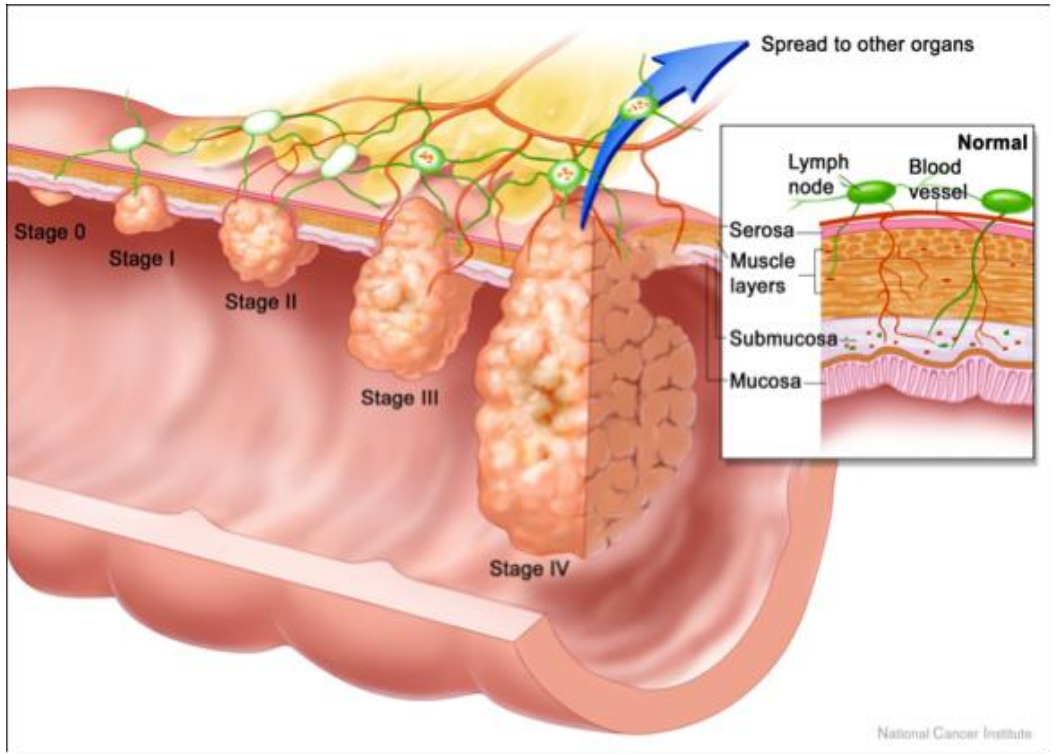
Invade surrounding tissue and establishment of secondary areas of growth: **Metastasis**



# Predictors of Behavior

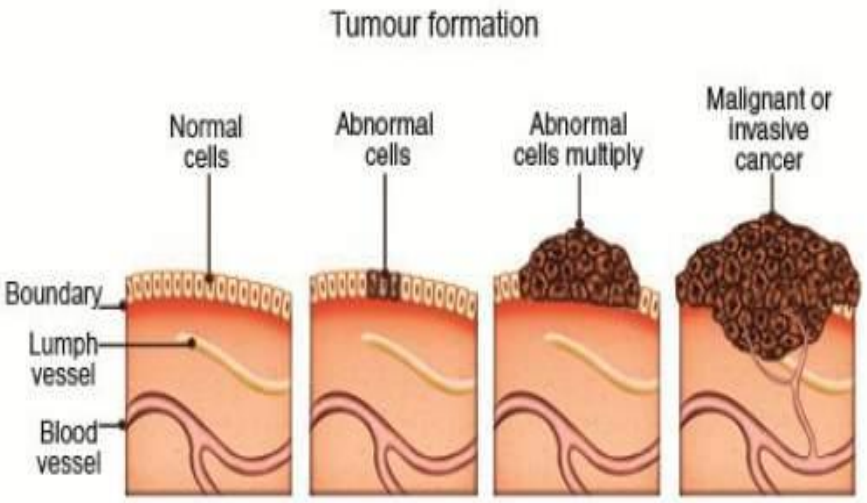
- **Grade – How bad do the cells look?**
- The pathologist gives the cancer a grade based on how different they look from normal cells (differentiation), how quickly they are growing and dividing, and how likely they are to spread. (How aggressive is the cancer?)
- **Stage – Where has the cancer spread?**
  - **Tumor (Size of Tumor)**
  - **Nodes (degree of spread to regional lymph nodes)**
  - **Metastasis (if cancer has spread to other body parts)**

Staging is a way of describing or classifying a cancer based on the extent of cancer in the body. The stage is often based on the size of the tumour, whether the cancer has spread (metastasized) from where it started to other parts of the body and where it has spread.



# Tumor size

| TUMOR (T) | DESCRIPTION |
|---|---|
| TX | Primary tumor cannot be evaluated |
| T0 | No evidence of primary tumor |
| Tis | Carcinoma in situ |
| T1 | Tumor < 2cm |
| T2 | Tumor 2-5cm |
| T3 | Tumor > 5cm |



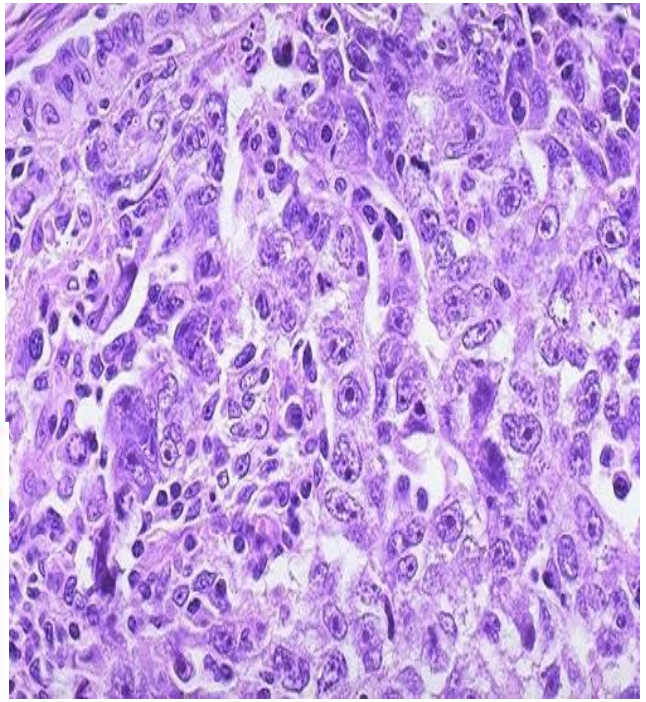

# Lymph Node Status

| LYMPH NOTE STATUS (N) | DESCRIPTION |
|---|---|
| NX | Regional lymph nodes cannot be evaluated |
| N0 | No regional lymph node involvement |
| N1 | 3 Lymph nodes + Axillary |
| N2 | 10 Lymph nodes+ |

# Distant Metastasis (+/-)

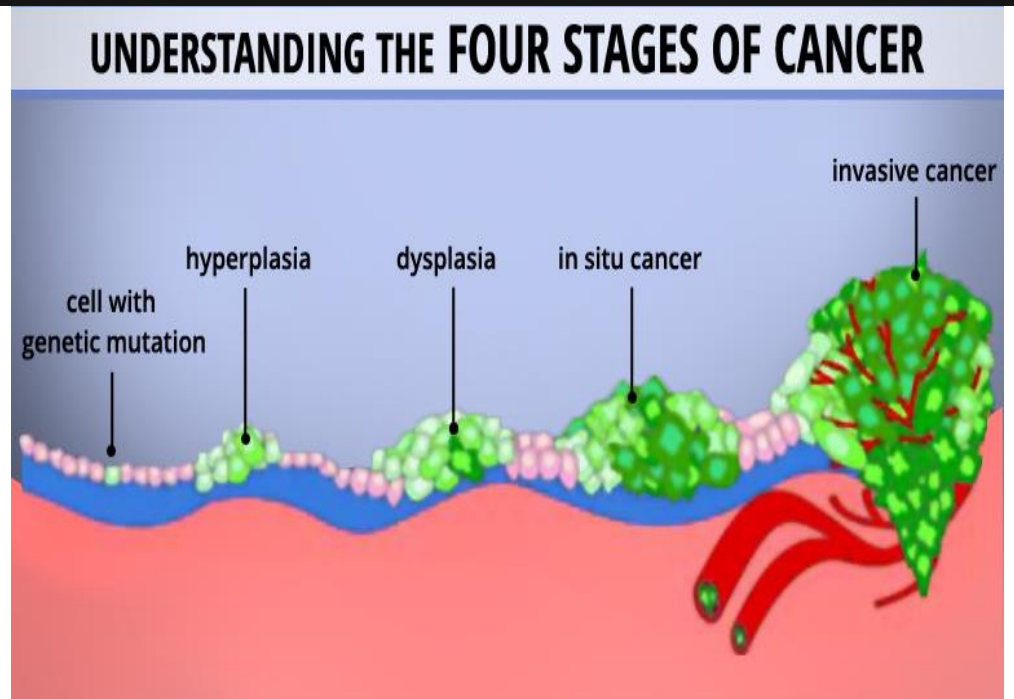
| METASTASIS | DESCRIPTION |
|---|---|
| MX | Distant metastasis cannot be evaluated |
| M0 | No distant metastasis |
| M1 | Distant metastasis is present |

# Grading Cancer



| GRADES | DESCRIPTION |
|---|---|
| GX | Grade cannot be assessed |
| G1 | Well differentiated (Low grade) |
| G2 | Moderately differentiated (Intermediate grade) |
| G3 | Poorly differentiated (High grade) |
| G4 | Undifferentiated (High grade-ANAPLASIA) |

| TNM Classification (American Joint Commission on Cancer) | | | | Dukes' Classification |
|---|---|---|---|---|
| Stages | T | N | M | Stages |
| Stage 0 | Tis | N0 | M0 | |
| Stage I | T1 | N0 | M0 | A |
| | T2 | N0 | M0 | B1 |
| Stage II | T3 | N0 | M0 | B2 |
| | T4 | N0 | M0 | B2 |
| Stage III | T1, T2 | N1 or N2 | M0 | C1 |
| | T3, T4 | N1 or N2 | M0 | C2 |
| Stage IV | Any T | Any N | M1 | D |



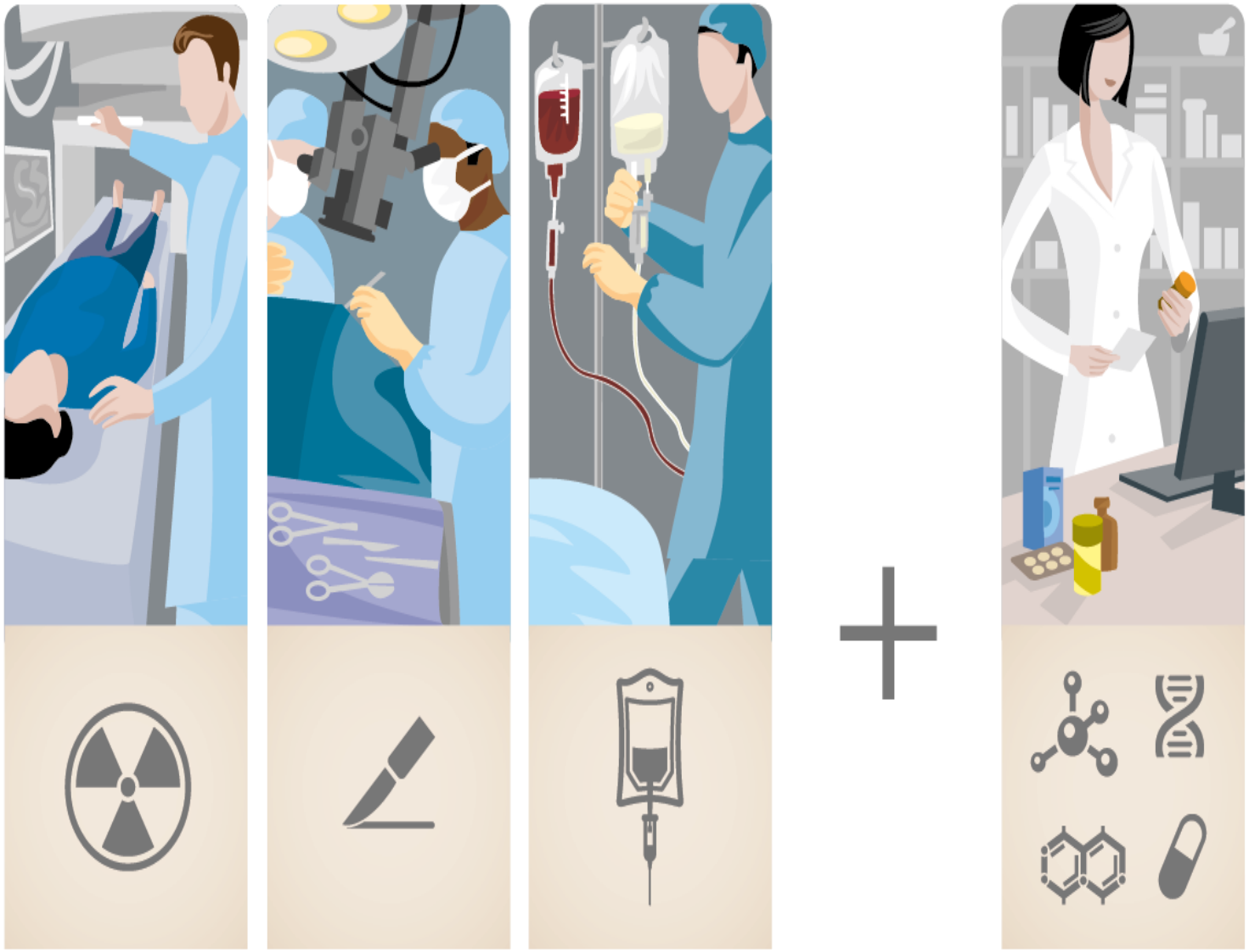
# What causes the mutations that lead to cancer?

- **Chemical Exposure**
  - Tobacco smoke
  - Environmental (PCBs)
  - Occupational (coal tar, asbestos, aniline dye)
  - Diet (aflatoxin)
- **Radiation (UV, ionizing)**
- **Infection**
  - Viruses (EBV, hepatitis B, papilloma)
  - Bacteria (Helicobacter)
- **Inherited familial cancer syndromes**

**What do these agents have in common?**

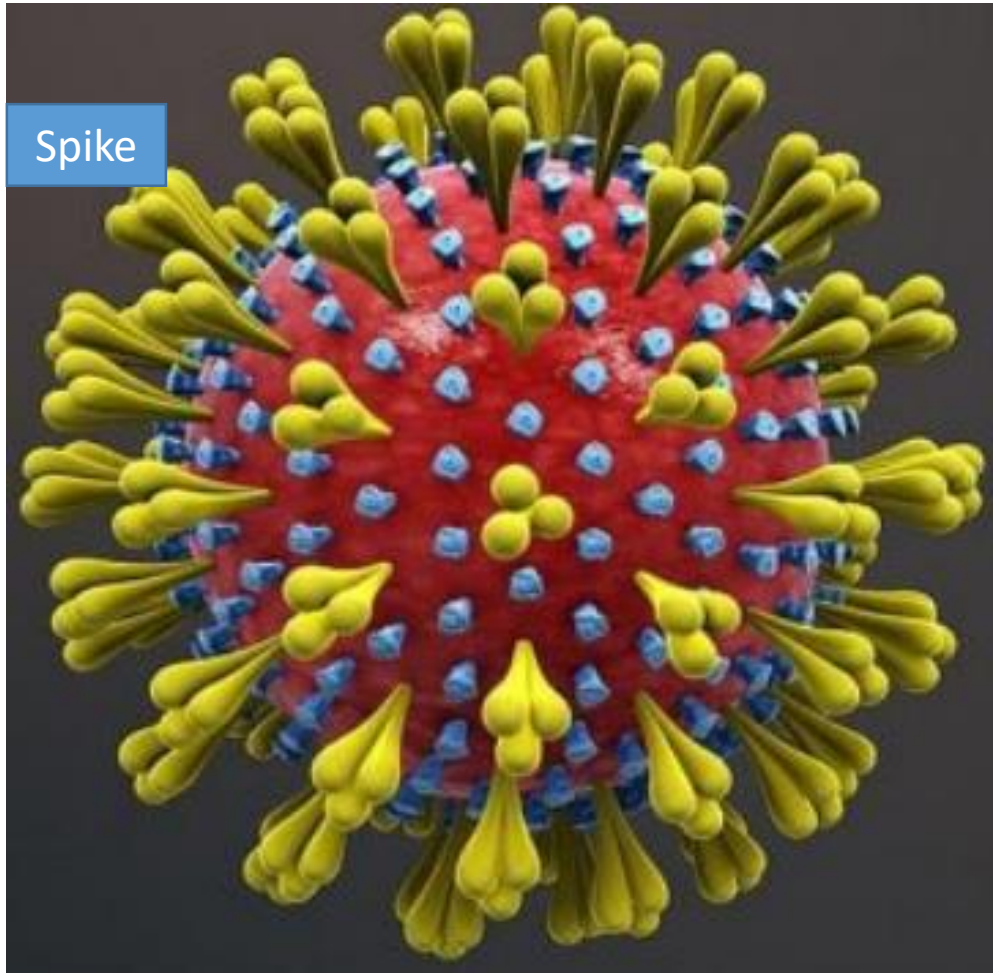
**Viruses:** insertional mutagenesis
**Chemicals:** DNA adducts
**UV and ionizing radiation:** single and double strand DNA breaks

# Traditional cancer treatment



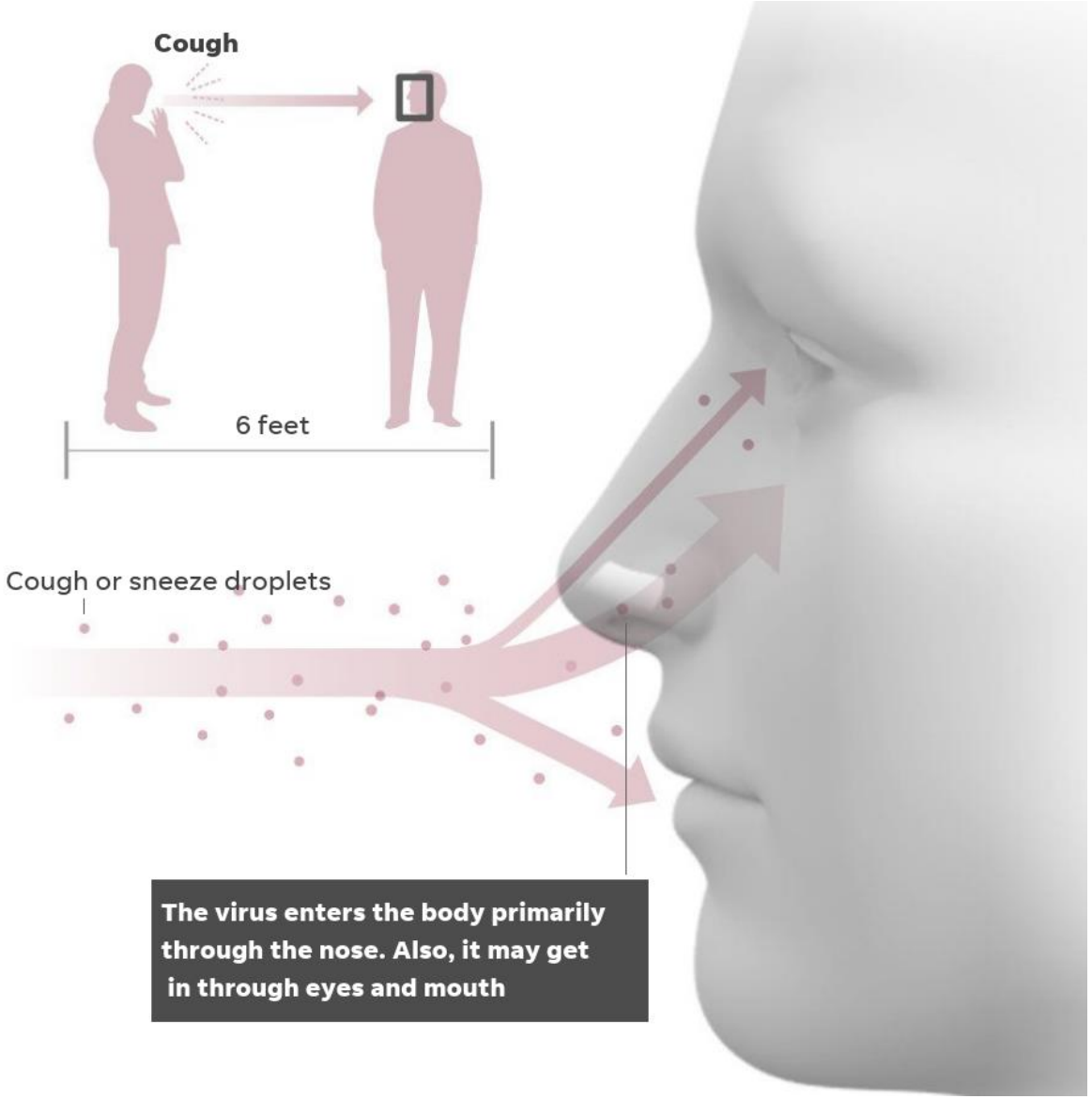
RADIATION SURGERY CHEMOTHERAPY + TARGETED DRUG THERAPIES

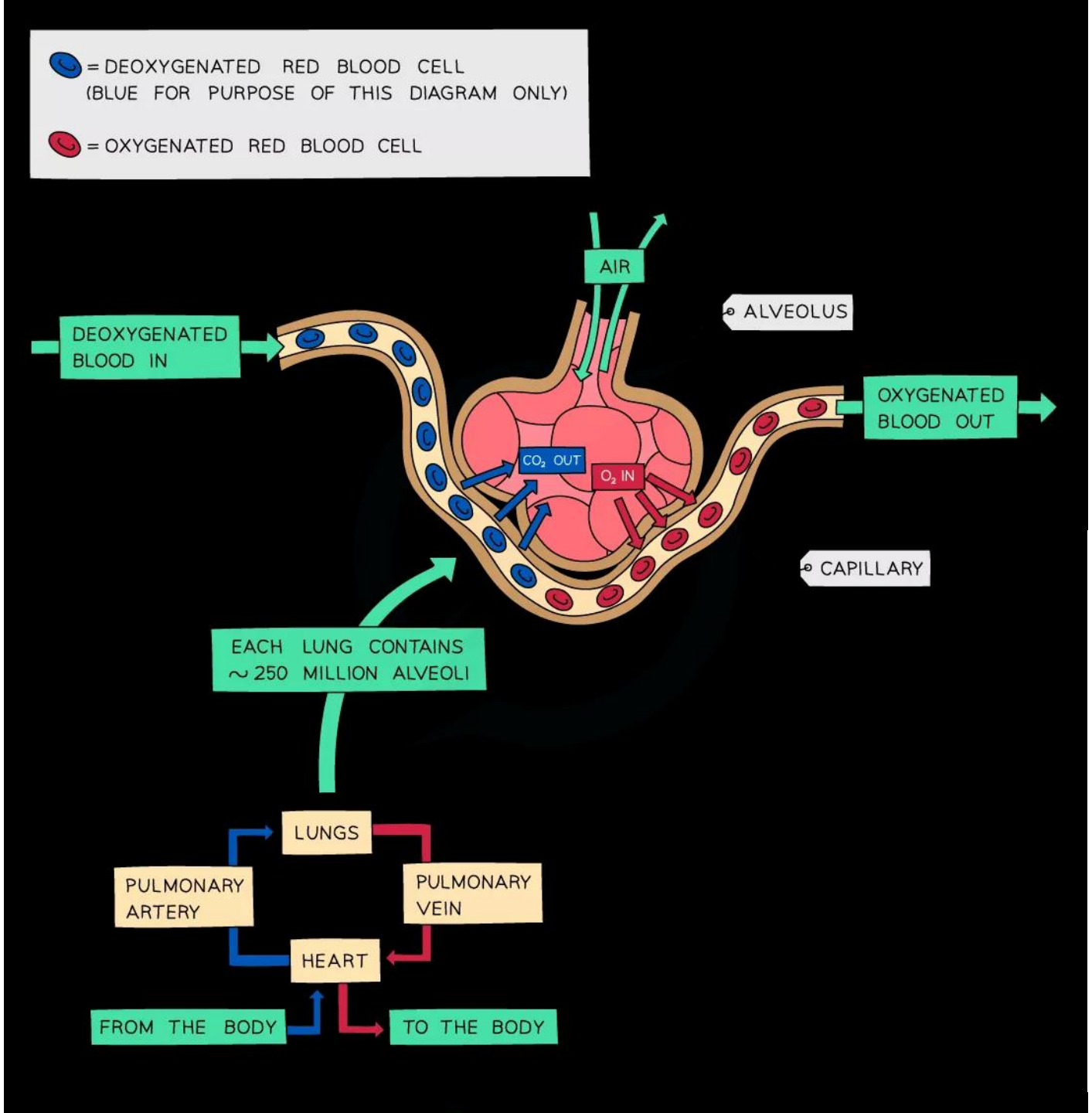
# Corona virus disease-2019 (COVID-19)

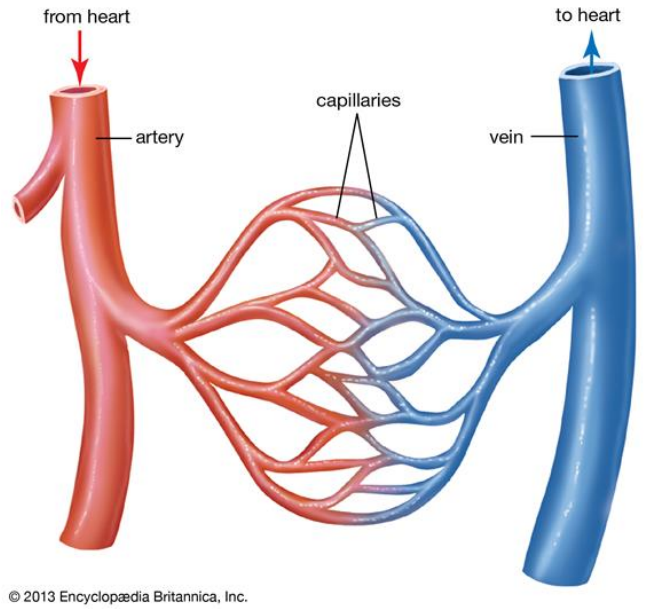



**Coronavirus**

**Name:** Severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2)




= DEOXYGENATED RED BLOOD CELL
(BLUE FOR PURPOSE OF THIS DIAGRAM ONLY)
= OXYGENATED RED BLOOD CELL
AIR
ALVEOLUS
DEOXYGENATED BLOOD IN
OXYGENATED BLOOD OUT
$CO_2$ OUT
$O_2$ IN
CAPILLARY
EACH LUNG CONTAINS ~250 MILLION ALVEOLI
LUNGS
PULMONARY ARTERY
PULMONARY VEIN
HEART
FROM THE BODY
TO THE BODY


from heart
to heart
capillaries
artery
vein
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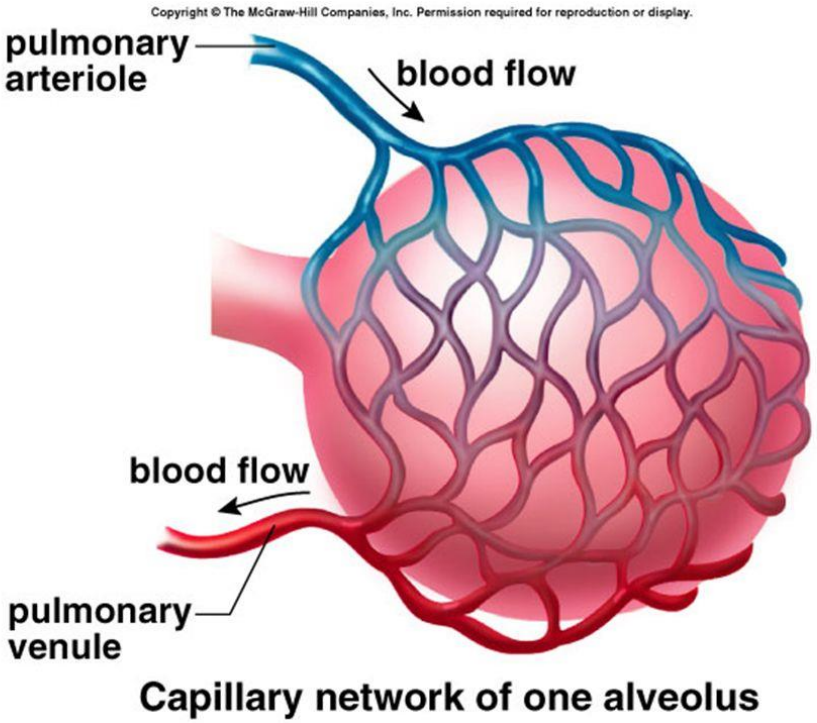
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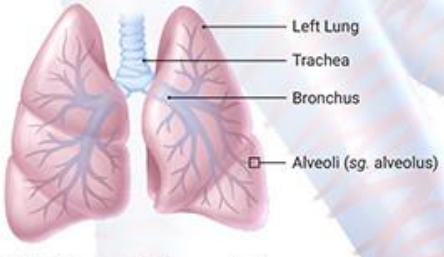
pulmonary arteriole
blood flow
blood flow
pulmonary venule
Capillary network of one alveolus

# COVID-19

## HOW DOES IT AFFECT YOU?

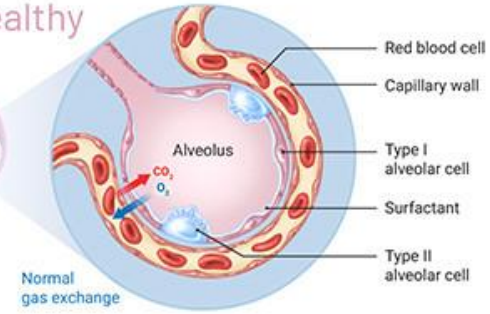
Designed by Avesta Rastan
www.azuravesta.com
@azuravesta
@azuraviz

Coronavirus Disease 2019 (COVID-19) is a pandemic caused by Severe Acute Respiratory Syndrome Coronavirus 2, also called SARS-CoV-2. Despite the widespread awareness regarding COVID-19, many are still unaware about how it affects the human body.



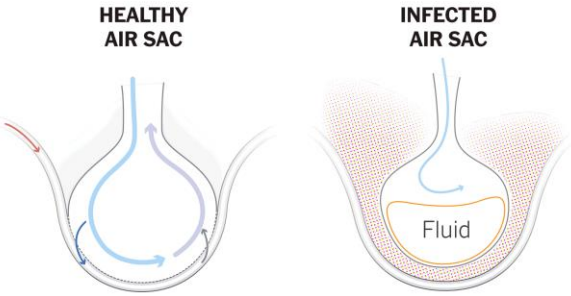

SARS-CoV-2 starts its journey in the nose, mouth, or eyes and travels down to the alveoli in the lungs. Alveoli are tiny sacs of air where gas exchange occurs.

## Healthy



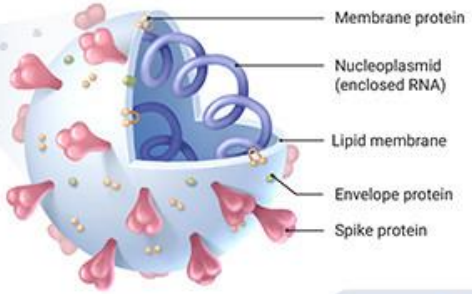

### Gas Exchange

Each sac of air, or alveolus, is wrapped with capillaries where red blood cells release **carbon dioxide** ($CO_2$) and pick up **oxyge** ($O_2$). Two alveolar cells facilitat gas exchange; **Type I** cells are thin enough that the oxygen passes right through, and **Type** cells secrete **surfactant** – a subtsance that lines the alveolu and prevents it from collapsing.




## Infected

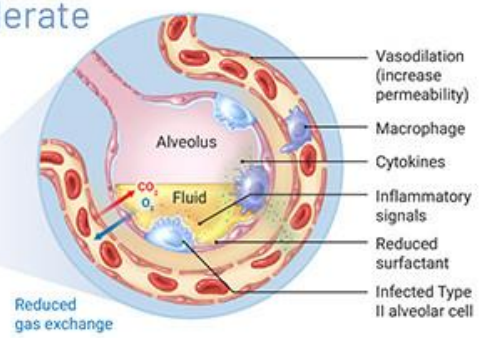
### SARS-CoV-2 Structure



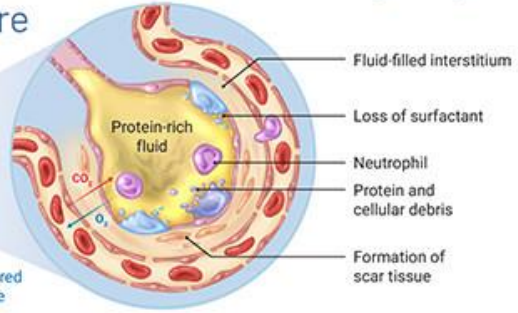

### Viral Infection

The spike proteins covering the coronavirus bind ACE2 receptors primarily on type II alveolar cells, allowing the virus to inject its RNA. The RNA "hijacks" the cell, telling it to assemble many more copies of the virus and release them into the alveolus. The host cell is destroyed in this process and the new coronaviruses infect neighbouring cells.

## Moderate




## Severe




### Impaired Gas Exchange

When the immune system attacks the area of infection it also kills healthy alveolar cells. This results in three things that hinder gas exchange:

1) Alveolar collapse due to loss of surfactant from Type II cells
2) Less oxygen enters the blood-stream due to lack of Type I cells
3) More fluid enters the alveolus

Stay Home

Symptoms may start to show (e.g. dry cough, fever, etc.)

Pneumonia develops

Shortness of breath

Hospitalization

Dangerous for high-risk individuals; secondary infections may occur

Intensive Care (ICU)

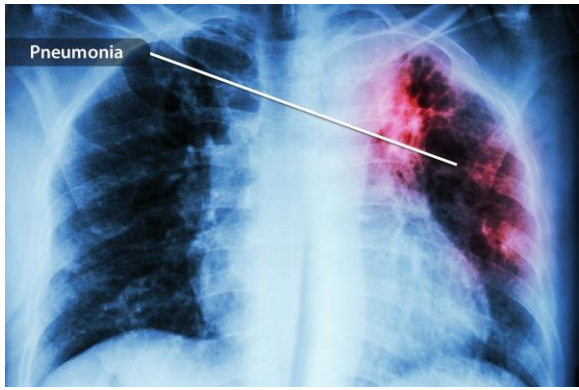
Patients may require ventilators and life-support

Complications unrelated to COVID-19 may occur

With proper care, patients may recover at any point during this process

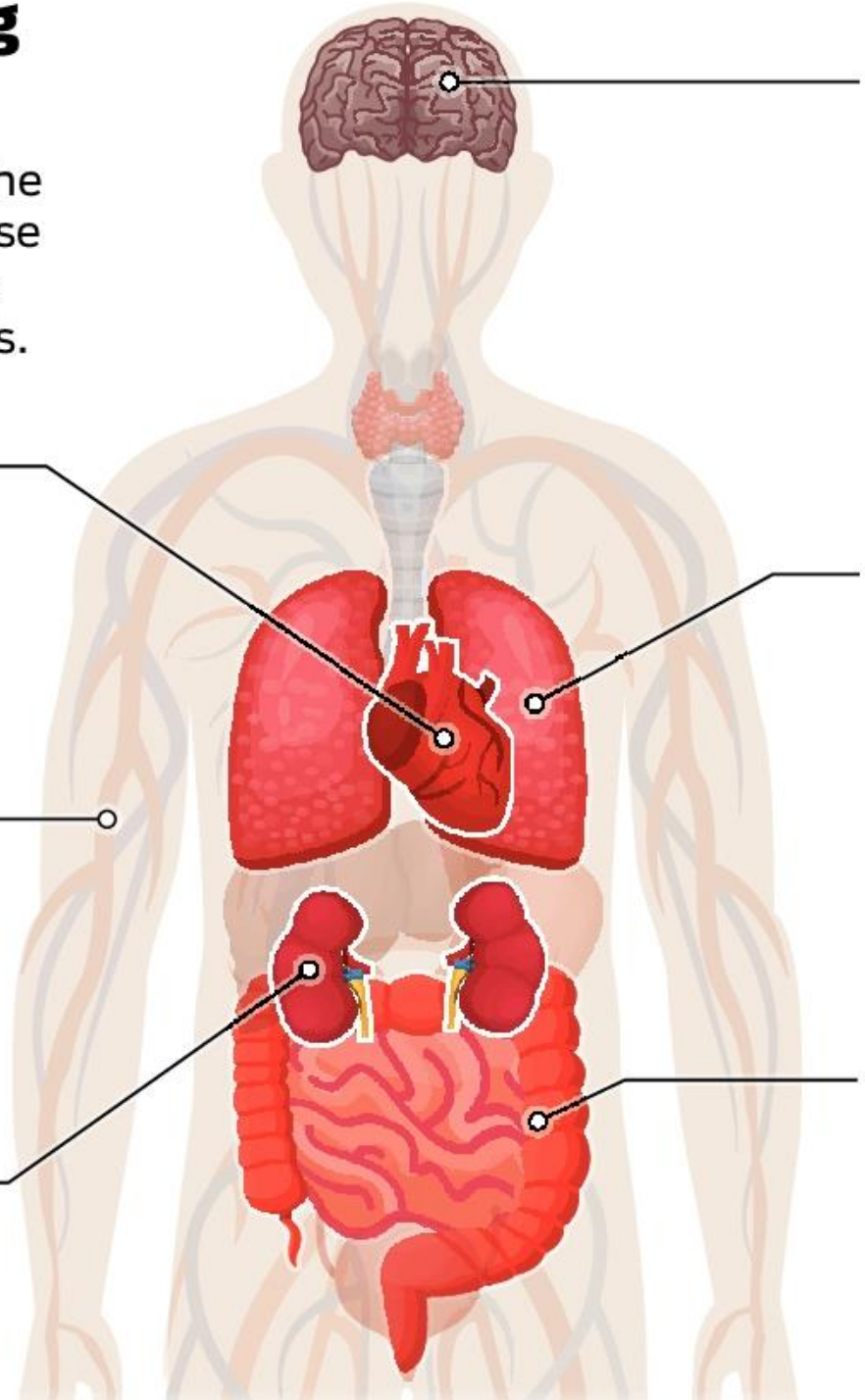
### Immune Response

1. After infection, Type II cells release **inflammatory signals** that recruit **macrophages** (immune cells).
2. Macrophages release **cytokines** that cause vasodilation, which allows more immune cells to come to the site of injury and exit the capillary.
3. Fluid accumulates inside the alveolus.
4. The fluid dilutes the surfactant which triggers the onset of alveolar collapse, decreasing gas exchange and increasing the work of breathing.
5. **Neutrophils** are recruited to the site of infection and release Reactive Oxygen Species (ROS) to destroy infected cells.
6. Type I and II cells are destroyed, leading to the collapse of the alveolus and causing **Acute Respiratory Distress Syndrome** (ARDS).
7. If inflammation becomes severe, the protein-rich fluid can enter the bloodstream and travel elsewhere in the body, causing **Systemic Inflammatory Response Syndrome** (SIRS).
8. SIRS may lead to **septic shock** and **multi-organ failure**, which can have fatal consequences.



# COVID-19's damaging effects on the body

Growing evidence suggests that the coronavirus, mostly known to cause respiratory illness, can also affect many of the body's primary organs.



**Brain**
People with COVID-19 have had strokes and seizures. Some have reported confusion or delirium. Not directly involving the brain but a central nervous issue: Many patients have reported losing their sense of smell.

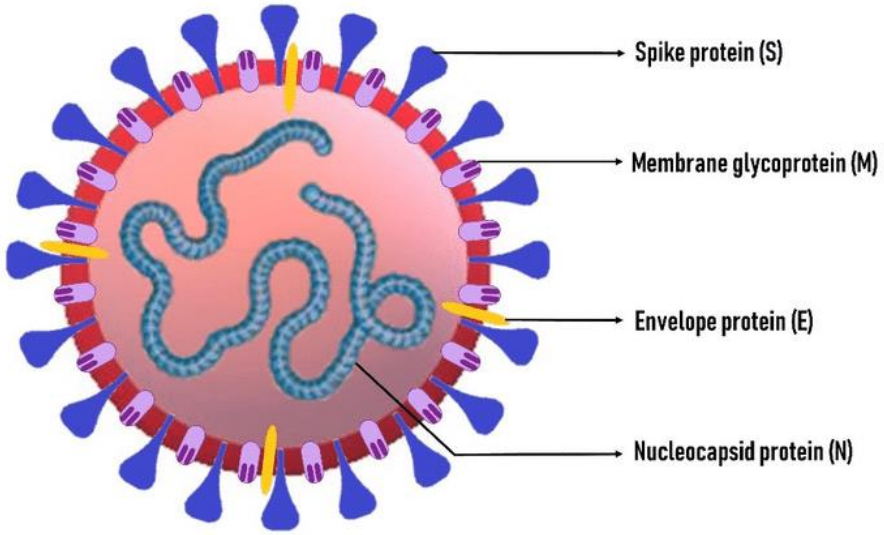
**Heart**
Doctors have reported inflammation to the heart and damage to the muscle. Some patients have died from severe heart attacks.

**Lungs**
The virus can cause pneumonia, in which the lungs become inflamed and fill with fluid. Patients may require ventilation. As the infection progresses, the virus can cause serious lung damage, which can be fatal.

**Blood vessels**
Blood clotting in major arteries and veins has been reported. Clots can break off and damage multiple organs by stopping blood flow.

**Kidneys**
Many COVID-19 patients suffer serious kidney damage and require dialysis.

**Intestines**
Roughly 20% of patients report diarrhea as an early symptom. The virus has been found in the lower intestinal tract of some patients.




# How do different Covid-19 vaccines work?

wellcome

## Viral vector

Uses a harmless virus which is altered to contain part of Covid-19's genetic code

## RNA (nucleic acid)

Contains a synthetic version of part of Covid-19's genetic code (messenger RNA)

## 'Whole' virus

Contains a weakened or inactivated version of the Covid-19 virus

## Protein subunit

Uses pieces of the Covid-19 virus - sometimes fragments of the 'spike' protein

The code tells our cells to make the Covid-19 'spike' protein, which triggers an immune response

This triggers an immune response

# COVID VACCINES: PEER COMPARISON

| VACCINE | SPUTNIK V | COVISHIELD | COVAXIN |
|---|---|---|---|
| DEVELOPED BY | RDIF | ASTRAZENECA | BHARAT BIOTECH |
| INDIAN MANUFACTURER | DRL, MANKIND, HETERO | SERUM INSTITUTE OF INDIA | BHARAT BIOTECH |
| TECHNOLOGY | USES TWO MODIFIED HUMAN ADENOVIRUS VECTOR | USES CHIMPANZEE ADENOVIRUS VECTOR | INACTIVATED VACCINE USING THE VIRUS STRAIN |